

AeroTwin — Market Research & Business Case

Industry Analysis, Competitive Landscape & Commercial Opportunity | Team FELONS | InnoVent-27

MARKET EXECUTIVE SNAPSHOT

\$81.9B	\$33B+	\$4.5B	22.4%
Global MRO Market (2023)	Engine MRO Segment (40% of total)	Predictive Maintenance Aviation Market (2026E)	Predictive Maint. CAGR (2023–2028)

THE AVIATION MRO INDUSTRY PROBLEM

► The Scale of the Challenge

Aviation is one of the world's most safety-critical and capital-intensive industries. A single commercial aircraft represents a capital asset worth \$50M–\$400M depending on type. The engines alone account for 30–40% of that value — with a GE90 or Trent XWB turbofan engine costing \$20–\$35M per unit. Maintaining these assets in airworthy condition is a continuous, complex, and extremely expensive operational challenge.

► Current Maintenance Models and Their Failures

The commercial aviation industry operates primarily under two maintenance paradigms, both of which have fundamental structural weaknesses:

Maintenance Model	How It Works	Critical Weaknesses
Fixed-Schedule Maintenance (FSM)	Aircraft are grounded and inspected on a predetermined calendar (e.g., every 500 flight hours or every 3 months, whichever comes first). The schedule is set conservatively — to be safe at the worst-case rate of degradation.	Over-maintenance: Components are often replaced or inspected when 40–60% of their useful life remains. Gartner estimates 30% of MRO costs are wasted on premature maintenance. Under-maintenance: When a component degrades faster than the conservative model expects, faults can develop undetected between inspection windows.
Reactive Maintenance	Equipment is operated until a fault triggers a warning or causes a failure.	Catastrophic risk: In aviation, reactive maintenance is unacceptable for flight-

	Maintenance is then performed reactively in response.	critical systems. An AOG (Aircraft on Ground) event triggered by an in-service fault costs \$150,000–\$300,000 per day in lost revenue, emergency engineering, and logistics — before safety-related costs. A major AOG event can cost \$10M+ for a wide-body long-haul aircraft.
Predictive Maintenance (Emerging — AeroTwin's target)	AI and sensor data are used to continuously monitor the actual health of components and predict when maintenance will be needed — based on measured degradation, not a conservative schedule.	Current barrier to adoption: Existing solutions (GE Aviation, Rolls-Royce) require proprietary hardware, massive fleet datasets, and multi-million dollar licensing — inaccessible to 60%+ of the MRO market.

► Quantified Cost of the Problem

Cost Category	Estimated Annual Cost	Source / Basis
Unnecessary scheduled maintenance (over-maintenance)	\$15–25B globally	30% of \$81.9B MRO market — Gartner / IATA estimates
Unplanned AOG events (reactive / missed faults)	\$8–12B globally	\$150K–\$300K per AOG day × ~40,000 AOG events/yr industry-wide
Premature component replacement	\$6–10B globally	Components replaced at 50–60% of true useful life on conservative FSM schedules
Safety incidents from missed gradual faults	Incalculable	1 in 12 aviation incidents linked to maintenance-related failures (IATA Safety Report)
Total addressable inefficiency	~\$30–47B per year	Conservative sum across above categories — the opportunity AeroTwin targets

MARKET SIZING — TAM / SAM / SOM

AeroTwin addresses a large, growing, and underserved segment of the aviation predictive maintenance market. We define our addressable market using the TAM / SAM / SOM framework:

Market Level	Definition	Size (2026E)	Rationale
TAM — Total Addressable Market	All aviation MRO spending globally, across all carriers, all aircraft types, all maintenance categories.	\$95B+	Global MRO market growing at 6.2% CAGR from \$81.9B (2023) baseline.
SAM — Serviceable	Engine MRO specifically (the segment AeroTwin targets),	\$38–42B	Engine MRO = ~40% of total MRO spend. Includes all

Addressable Market	across all carriers with turbo-fan powered aircraft.		scheduled + unscheduled engine maintenance.
SOM — Serviceable Obtainable Market	Regional airlines, charter carriers, third-party MRO providers, and smaller national carriers currently underserved by enterprise Digital Twin solutions.	\$4–8B	60%+ of the MRO market by number of operators — currently priced out of Rolls-Royce/GE solutions. AeroTwin's open, low-cost architecture targets this segment.

📈 MARKET GROWTH DRIVER

The global aviation predictive maintenance market is growing at 22.4% CAGR (2023–2028), driven by rising fleet sizes, increasing regulatory pressure for condition-based maintenance documentation, and the falling cost of edge AI compute hardware (NVIDIA Jetson, Coral TPU). AeroTwin is entering this market at exactly the right inflection point.

COMPETITIVE LANDSCAPE

► Tier 1 Competitors — Enterprise OEM Solutions

These are the industry-leading Digital Twin platforms that AeroTwin benchmarks against:

Company / Product	Target Customer	Core Capability	Deployment Cost	Hardware Dependency	SME Accessible?
Rolls-Royce IntelligentEngine	Large carriers with RR engines only	Full DT with satellite telemetry, 5,000+ parameters per engine	\$\$\$\$ (Tens of millions per fleet)	Proprietary — embedded at manufacture	✗ No
GE Aviation Digital Twin Platform	Large carriers with GE/CFM engines	Engine performance monitoring + fuel optimisation + RUL	\$\$\$\$ (Enterprise licensing)	Proprietary — embedded at manufacture	✗ No
Boeing AnalytX	Airlines operating Boeing fleet	Structural + engine health, FOQA data integration	\$\$\$ (Included with Boeing services contracts)	Proprietary — Boeing airframe sensors	✗ No
Airbus SKYWISE	Airlines operating Airbus fleet	Fleet health monitoring, predictive alerts, MRO scheduling	\$\$\$ (SaaS model, still enterprise pricing)	Airbus sensor ecosystem	✗ No
AeroTwin (Team FELONS)	Regional airlines, 3rd-party MROs, smaller carriers	Real-time DT, RUL prediction, anomaly detection, open	\$ (Near-zero hardware cost — open source)	None (simulation) → Arduino → Jetson	✔Yes — by design

architecture

► AeroTwin's Competitive Differentiation — Porter's 5 Forces View

AeroTwin is differentiated along five dimensions that the Tier 1 competitors cannot match:

- ◆ **Cost Accessibility:** Zero proprietary hardware dependency and open-source software stack means AeroTwin can be deployed at near-zero infrastructure cost — versus tens of millions for enterprise OEM solutions. This makes predictive Digital Twin technology accessible to the 60%+ of the MRO market that Rolls-Royce and GE cannot reach.
- ◆ **Hardware Independence:** AeroTwin is not locked to any aircraft manufacturer's sensor ecosystem. It is designed to work with any sensor source — from a ₹200 Arduino MEMS sensor to an industrial Jetson-based edge compute unit — through a single configurable data ingestion interface.
- ◆ **Transparency & Explainability:** Unlike black-box enterprise platforms, AeroTwin uses an interpretable Random Forest model whose feature importance rankings can be displayed on the dashboard. Maintenance engineers can see not just 'RUL is 280 hours' but 'because vibration rate-of-change is the top contributor to this prediction.' This builds trust and supports regulatory compliance.
- ◆ **Open Architecture for Customisation:** AeroTwin's modular design allows any MRO operator to replace or extend individual components — swap the fatigue formula, add a new sensor type, change the alert thresholds — without rewriting the system. This is fundamentally impossible with locked enterprise OEM platforms.
- ◆ **Educational & Research Value:** As a transparent, open-source implementation of Digital Twin principles calibrated against NASA's C-MAPSS benchmark, AeroTwin has dual-use value: as a commercial predictive maintenance tool and as an educational platform for training next-generation aircraft maintenance engineers — a market that no enterprise OEM platform addresses at all.

BUSINESS MODEL & REVENUE STRATEGY

► Proposed Commercial Model — SaaS + Professional Services

If AeroTwin were to be commercialised beyond the hackathon prototype stage, the following revenue model is proposed based on comparable SaaS predictive maintenance platforms in the industrial IoT space:

Revenue Stream	Model	Indicative Pricing	Target Customer
Tier 1 — Starter SaaS	Cloud-hosted AeroTwin dashboard. Simulation-based (no hardware). Up to 5 engines monitored.	\$500/month per operator	Small charter operators, flight schools, MRO training academies
Tier 2 — Professional SaaS	Self-hosted or cloud. Real IoT sensor integration (Arduino/Jetson). Up to 50 engines. Priority support.	\$3,500/month per operator	Regional airlines (20–50 aircraft), third-party MRO providers
Tier 3 — Enterprise	Full deployment, custom integration with existing ACARS/FOQA systems,	Custom — \$50K–\$200K/year	Mid-size carriers (50–200 aircraft), national MRO hubs

	custom alert rules, dedicated SLA.		
Professional Services	Implementation consulting, custom sensor integration, staff training, data migration from paper logs.	\$150–\$250/hour	All tiers — especially Tier 2 and 3 deployments
Research & Education Licensing	Open-source core with commercial licence for educational institutions using AeroTwin for maintenance training simulations.	\$5,000–\$20,000/year per institution	Aviation training academies, university aerospace engineering programs

► Financial Impact Projections — Per Customer ROI

The following illustrates the ROI case for a typical Tier 2 customer (regional airline, 30 aircraft fleet):

Category	Current State (Without AeroTwin)	With AeroTwin (Estimated)
Annual maintenance inspections (30 aircraft × 4 engines each)	480 inspections/year on fixed schedule	~340 inspections/year — 30% reduction via RUL-based scheduling
Avg. cost per inspection (labour + downtime)	\$12,000 per inspection	Same per inspection — fewer total inspections needed
Annual inspection cost	\$5.76M/year	~\$4.08M/year — saving ~\$1.68M/year
Unplanned AOG events (industry avg: 2.5 per fleet per year)	~2.5 events × \$200K avg = \$500K/year	~1.5 events × \$200K = \$300K/year — 40% reduction via early detection
Component premature replacement (conservative schedule)	\$800K/year (est.)	\$560K/year — 30% reduction via true RUL-based replacement
AeroTwin Tier 2 SaaS cost	N/A	\$42,000/year (\$3,500/month)
NET ANNUAL SAVING	Baseline	~\$2.078M/year gross — ~\$2.036M/year net of AeroTwin cost
ROI on AeroTwin investment	N/A	~4,800% annual ROI — payback in under 1 week of operation

GO-TO-MARKET STRATEGY

► Phase 1 — Hackathon to Open-Source (Months 0–6)

Release AeroTwin as a fully documented open-source project on GitHub following InnoVent-27. Build community around the project by publishing the NASA C-MAPSS calibration methodology and inviting contributions from aerospace engineering students and MRO researchers globally. Open-source visibility creates credibility and a developer base before any commercial launch.

► Phase 2 — Pilot Program with Regional MRO Partner (Months 6–18)

Partner with one regional airline or third-party MRO operator (targeting Indian carriers such as SpiceJet MRO, Air India Engineering Services, or GMR Aero Technic) for a no-cost pilot deployment. Integrate AeroTwin with their existing FOQA data download workflow (replacing the simulation engine with real post-flight data). Build a case study with quantified outcomes for Series A fundraising.

► Phase 3 — Regulated Commercial Launch (Months 18–36)

Pursue DGCA (India) and EASA (Europe) software certification for AeroTwin as a health monitoring decision-support tool (not a safety-critical system — the distinction keeps regulatory burden manageable). Launch Tier 1 and Tier 2 SaaS products. Target 10 paying operators in the first year.

► Why India Is the Right First Market

- ◆ India is the world's fastest-growing aviation market — DGCA projects 300M+ passengers by 2030, requiring a doubling of the current fleet and a massive expansion of MRO capacity.
- ◆ The Indian government's MRO policy reforms (2020–2024) aim to grow India's MRO industry from \$1.7B to \$4B by 2030 — specifically targeting domestic predictive maintenance capability.
- ◆ IndiGo, Air India, and SpiceJet collectively operate 500+ aircraft, most without access to OEM-level Digital Twin monitoring — a ready, underserved first customer base.
- ◆ Tata Technologies itself is a major MRO ecosystem player through its parent group — an ideal strategic partner for AeroTwin's commercial launch.

RISK ANALYSIS & MITIGATION

Risk	Likelihood	Impact	Mitigation
Simulation data not representative of real engine behaviour	Medium	High	Calibrate simulator against NASA C-MAPSS and FEMTO datasets. Pilot program with real FOQA data validates and recalibrates model before commercial deployment.
Enterprise OEMs launch lower-cost products targeting SME market	Low	High	AeroTwin's hardware-independence and open architecture create a structural cost advantage that proprietary OEM products cannot replicate — they cannot give away their sensor ecosystem.
DGCA/EASA regulatory certification takes longer than expected	Medium	Medium	Position AeroTwin as a 'decision support tool' (non-safety-critical) — a significantly lighter regulatory pathway than a certified airworthiness

			system.
Airlines reluctant to adopt unproven student-developed software for operational use	High (initially)	Medium	Address through open-source release, academic validation, and a no-cost pilot program with full data ownership retained by the airline. Trust is built through transparency, not marketing.
ML model accuracy degrades on real data not seen in training	Medium	Medium	Implement online learning — the model retrains incrementally as real data accumulates during the pilot, improving accuracy over time through operational use.